Tugas Praktikum Pemrograman Mobile
Pertemuan Ke - 3



Nama : **Michael Cornelius Bong**
NIM : **20240801004**
Jurusan : **Ilmu Komputer**
Prodi : **Teknik Informatika**

Universitas Esa Unggul

Praktikum Flutter

1. Tujuan

Praktikum Ketiga bertujuan mempelajari cara mengambil input pengguna dengan TextField dan TextEditingController, membuat form dengan validasi, memahami keterbatasan setState() ketika data digunakan banyak halaman, menerapkan state management menggunakan ChangeNotifier dan Provider, serta mengenal perbedaan context.watch() dan context.read().

2. Alat dan Bahan

- Komputer Atau Laptop
- Flutter Package (Flutter - Install)
- Android Studio / IDE Dengan Support Flutter & Dart
- Emulator SDK Android / HP Android Dengan Debugging Mode

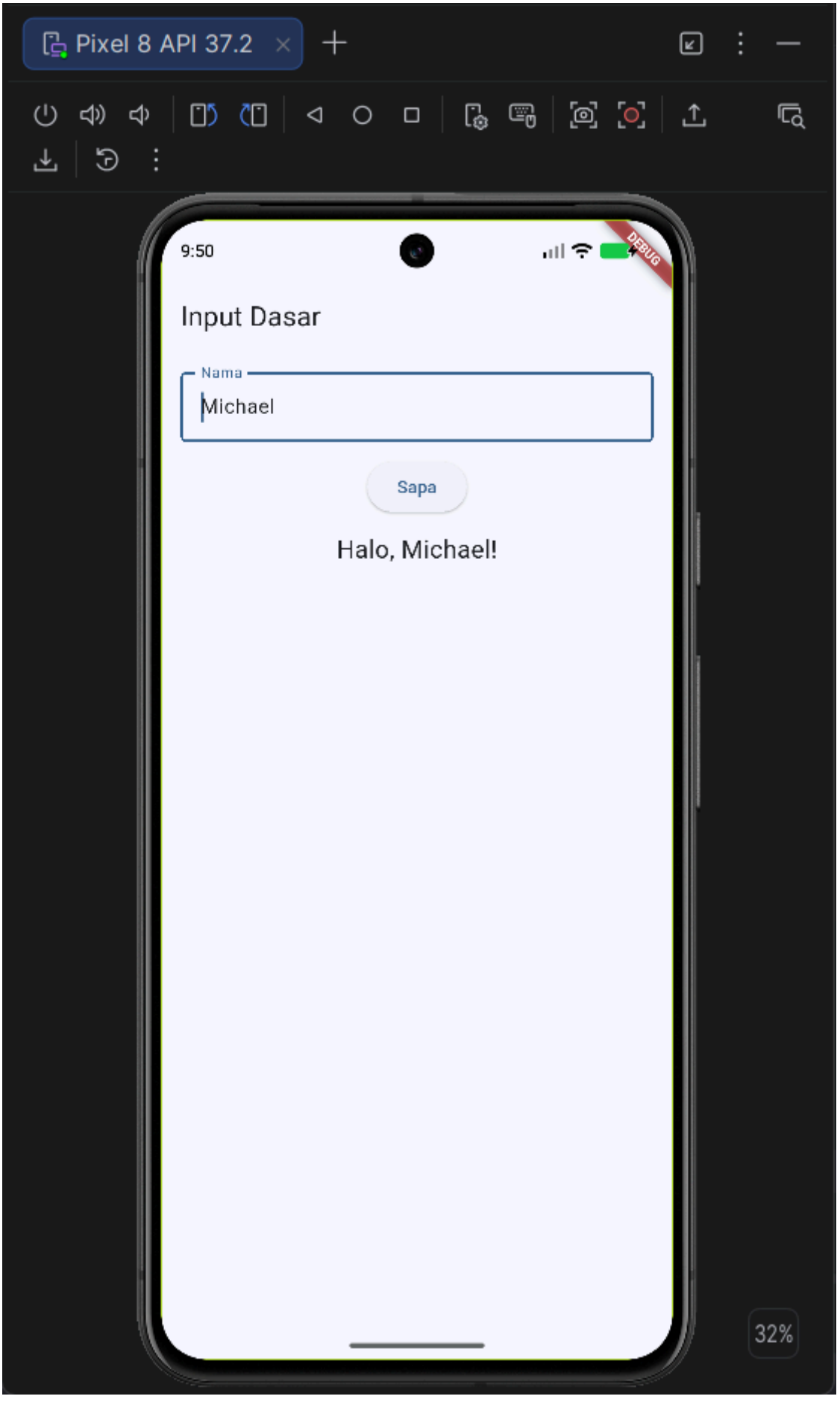
3. Pembuatan Tampilan Menginput Data

3.1 Input Dasar Dengan TextField :

main.dart

```dart
import 'package:flutter/material.dart';
void main() => runApp(const MyApp());
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Praktikum 3',
      theme: ThemeData(colorSchemeSeed: Colors.blue, useMaterial3: true),
      home: const InputPage(),
    );
  }
}
```

```dart
class InputPage extends StatefulWidget {
  const InputPage({super.key});
  @override
  State<InputPage> createState() => _InputPageState();
}
class _InputPageState extends State<InputPage> {
  final _controller = TextEditingController();
  String _salam = '';
  @override
  void dispose() {
    _controller.dispose();
    super.dispose();
  }
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Input Dasar')),
      body: Padding(
        padding: const EdgeInsets.all(16),
        child: Column(
          children: [
            TextField(
              controller: _controller,
              decoration: const InputDecoration(
                labelText: 'Nama',
                border: OutlineInputBorder(),
              ), InputDecoration
            ), TextField
            const SizedBox(height: 12),
            ElevatedButton(
              onPressed: () {
                setState(() => _salam = 'Halo, ${_controller.text}!');
              },
              child: const Text('Sapa'),
            ), ElevatedButton
            const SizedBox(height: 12),
            Text(_salam, style: const TextStyle(fontSize: 20)),
          ],
        ), Column
      ), Padding
    ); Scaffold
  }
}
```


Pixel 8 API 37.2
9:50
DEBUG
Input Dasar
Nama
Michael
Sapa
Halo, Michael!
32%

3.2 Pembuatan Form Dengan Adanya Validasi :

main.dart

```dart
import 'package:flutter/material.dart';

void main() => runApp(const MyApp());

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Praktikum 3',
      theme: ThemeData(colorSchemeSeed: Colors.blue, useMaterial3: true),
      home: const FormPage(),
    );
  }
}

class FormPage extends StatefulWidget {
  const FormPage({super.key});

  @override
  State<FormPage> createState() => _FormPageState();
}

class _FormPageState extends State<FormPage> {
  final _formKey = GlobalKey<FormState>();
  final _nama = TextEditingController();
  final _email = TextEditingController();
  String? _jurusan;
  bool _setuju = false;
```

```dart

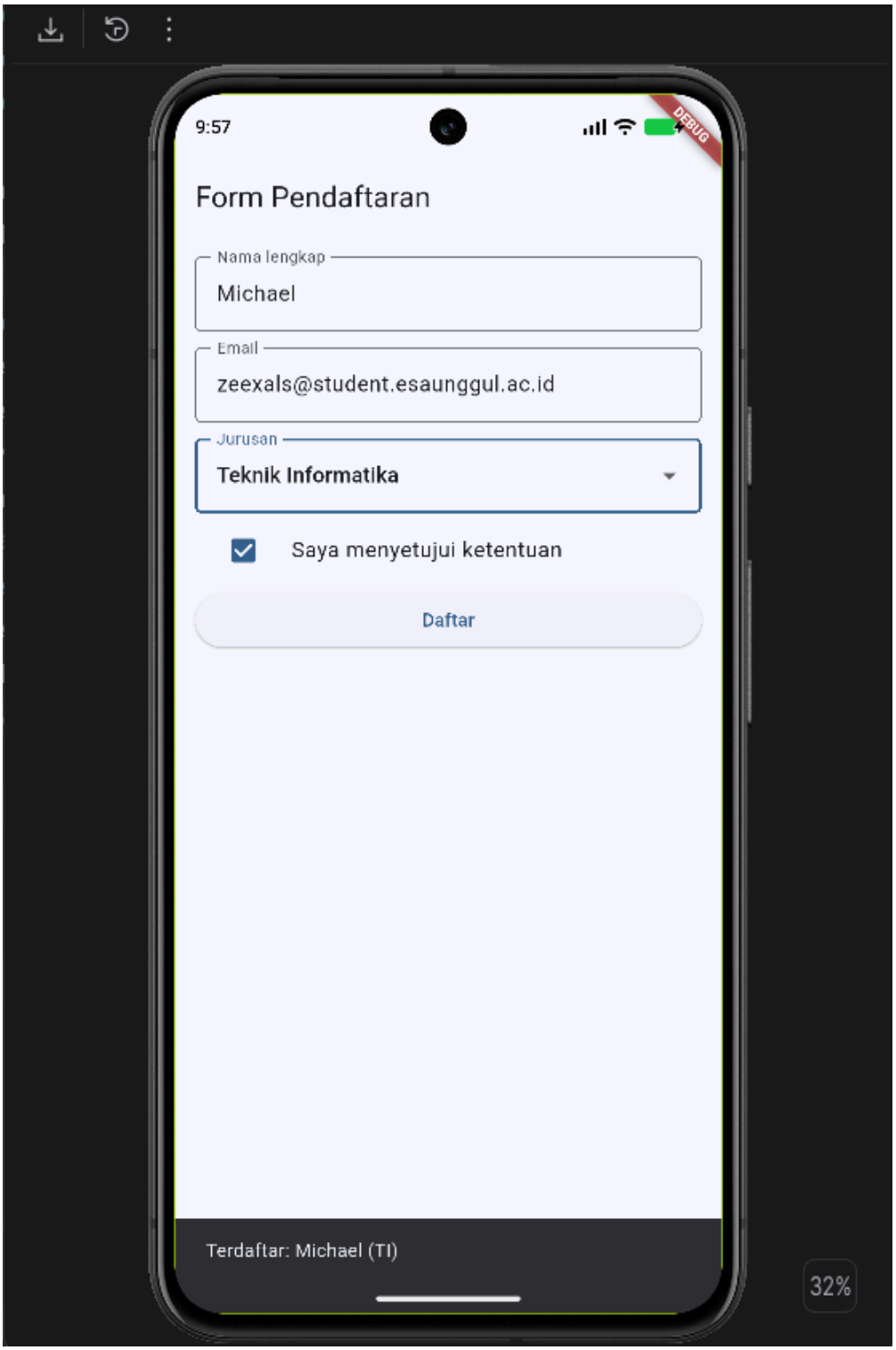
  @override
  void dispose() {
    _nama.dispose();
    _email.dispose();
    super.dispose();
  }

  void _kirim() {
    if (_formKey.currentState!.validate()) {
      final jurusan = _jurusan ?? '-';
      ScaffoldMessenger.of(context).showSnackBar(
        SnackBar(content: Text('Terdaftar: ${_nama.text} ($jurusan)')),
      );
    }
  }

```

```dart
@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(title: const Text('Form Pendaftaran')),
    body: Form(
      key: _formKey,
      child: ListView(
        padding: const EdgeInsets.all(16),
        children: [
          TextFormField(
            controller: _nama,
            decoration: const InputDecoration(
              labelText: 'Nama lengkap',
              border: OutlineInputBorder(),
            ), InputDecoration
            validator: (v) =>
            (v == null || v.trim().isEmpty) ? 'Nama wajib diisi' : null,
          ), TextFormField
          const SizedBox(height: 12),
          TextFormField(
            controller: _email,
            keyboardType: TextInputType.emailAddress,
            decoration: const InputDecoration(
              labelText: 'Email',
              border: OutlineInputBorder(),
            ), InputDecoration
            validator: (v) {
              if (v == null || !v.contains('@')) return 'Email tidak valid';
              return null;
            },
          ), TextFormField
          const SizedBox(height: 12),
          DropdownButtonFormField<String>(
            decoration: const InputDecoration(
              labelText: 'Jurusan',
              border: OutlineInputBorder(),
            ), InputDecoration
            items: const [
              DropdownMenuItem(value: 'TI', child: Text('Teknik Informatika')),
              DropdownMenuItem(value: 'SI', child: Text('Sistem Informasi')),
              DropdownMenuItem(value: 'TE', child: Text('Teknik Elektro')),
            ],
            onChanged: (v) => setState(() => _jurusan = v),
            validator: (v) => v == null ? 'Pilih jurusan' : null,
          ), DropdownButtonFormField
          CheckboxListTile(
            title: const Text('Saya menyetujui ketentuan'),
            value: _setuju,
            controlAffinity: ListTileControlAffinity.leading,
            onChanged: (v) => setState(() => _setuju = v ?? false),
```

```
98                ), CheckboxListTile
99                ElevatedButton(
100                 onPressed: _setuju ? _kirim : null,
101                 child: const Text('Daftar'),
102               ), ElevatedButton
103             ],
104           ), ListView
105         ), Form
106       ); Scaffold
107     }
108   }
```



3.3 Instalasi Provider dan Penggunaannya :

```
PS C:\Users\Asus\Documents\University\Semester 5\Pemrograman Mobile\Praktikum\praktikum_tiga> flutter pub add provider
Resolving dependencies...
Downloading packages...
  cupertino_icons 1.0.9 (2.0.0 available)
  material_color_utilities 0.13.0 (0.13.1 available)
  meta 1.18.3 (1.19.0 available)
+ nested 1.0.0
+ provider 6.1.5+1
  test_api 0.7.12 (0.7.14 available)
  vector_math 2.4.0 (2.4.3 available)
Changed 2 dependencies!
```

main.dart

```dart
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';

class Tugas {
  String judul;
  bool selesai;
  Tugas(this.judul, {this.selesai = false});
}
class TugasModel extends ChangeNotifier {
  final List<Tugas> _items = [];
  List<Tugas> get items => List.unmodifiable(_items);
  int get jumlahSelesai => _items.where((t) => t.selesai).length;
  void tambah(String judul) {
    _items.add(Tugas(judul));
    notifyListeners();
  }
  void toggle(int index) {
    _items[index].selesai = !_items[index].selesai;
    notifyListeners();
  }
  void hapus(int index) {
    _items.removeAt(index);
    notifyListeners();
  }
}

void main() {
  runApp(
    ChangeNotifierProvider(
      create: (_) => TugasModel(),
```

```dart
      child: const MyApp(),
    ), ChangeNotifierProvider
  );
}
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Daftar Tugas',
      theme: ThemeData(colorSchemeSeed: Colors.blue, useMaterial3: true),
      home: const TugasPage(),
    ); MaterialApp
  }
}

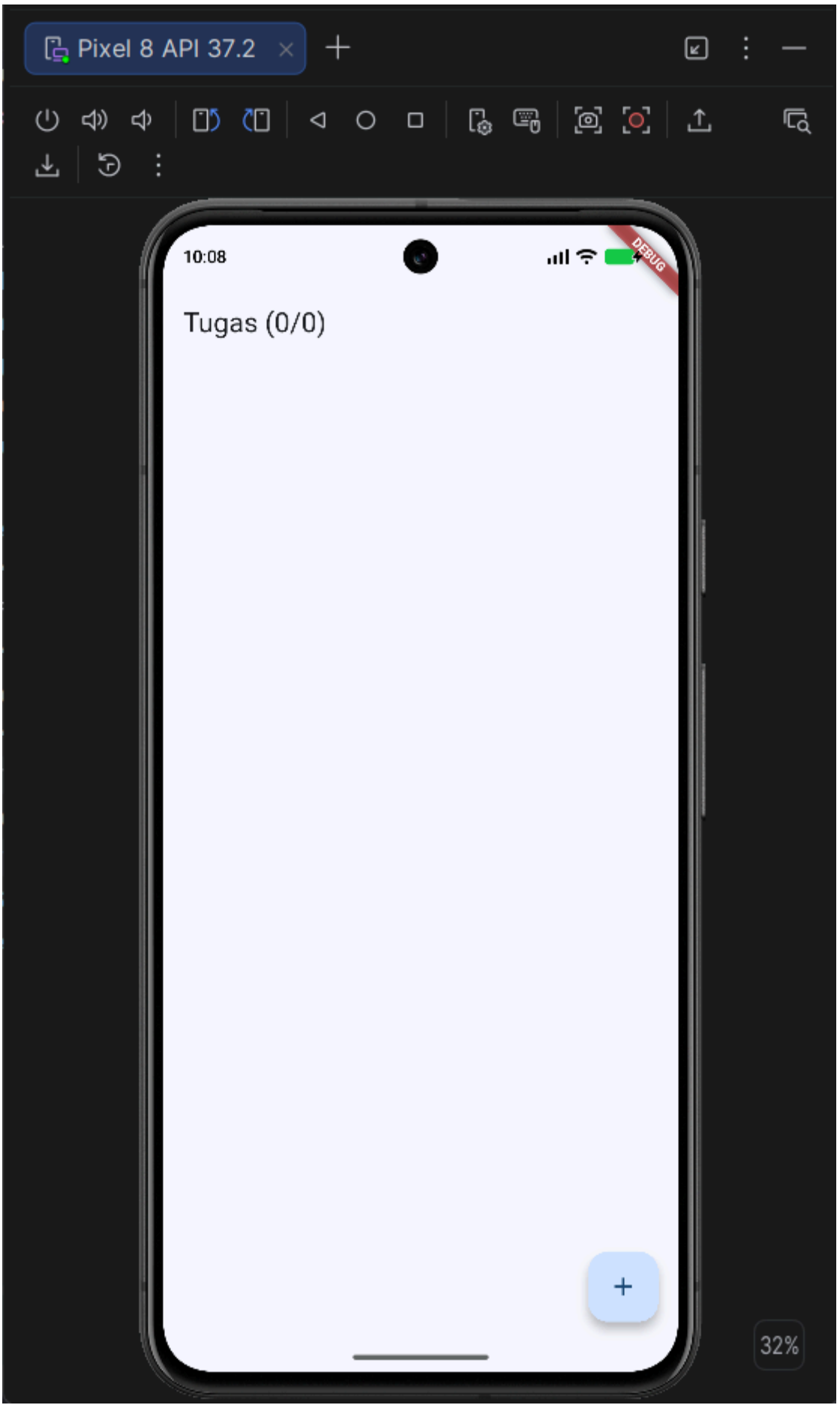
class TugasPage extends StatelessWidget {
  const TugasPage({super.key});
  @override
  Widget build(BuildContext context) {
    final model = context.watch<TugasModel>();
    return Scaffold(
      appBar: AppBar(
        title: Text('Tugas (${model.jumlahSelesai}/${model.items.length})'),
      ), AppBar
      body: ListView.builder(
        itemCount: model.items.length,
        itemBuilder: (context, i) {
          final t = model.items[i];
          return ListTile(
```

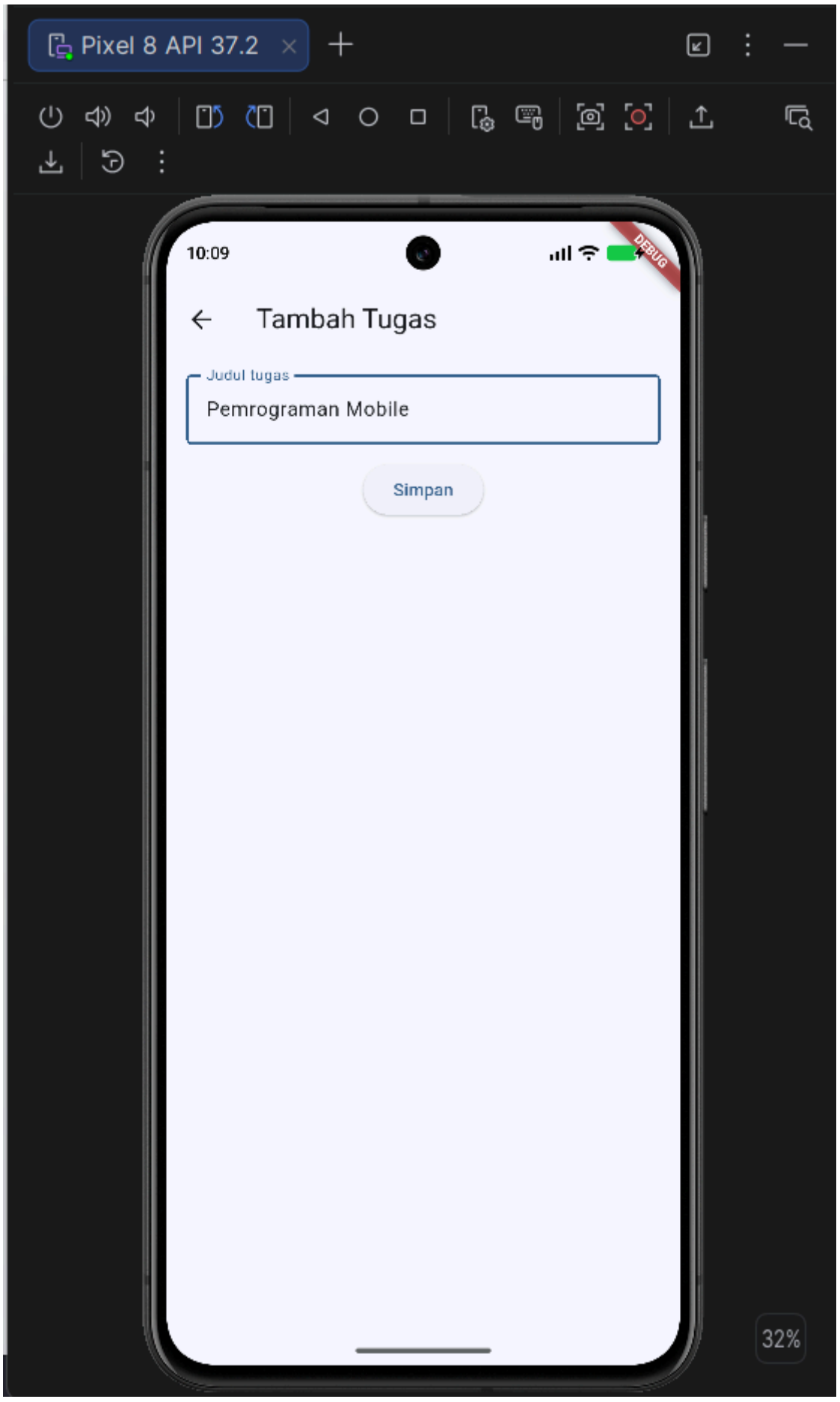
```dart
            leading: Checkbox(
              value: t.selesai,
              onChanged: (_) => context.read<TugasModel>().toggle(i),
            ), Checkbox
            title: Text(
              t.judul,
              style: TextStyle(
                decoration: t.selesai ? TextDecoration.lineThrough : null,
              ), TextStyle
            ), Text
            trailing: IconButton(
              icon: const Icon(Icons.delete),
              onPressed: () => context.read<TugasModel>().hapus(i),
            ), IconButton
          ); ListTile
        },
      ), ListView.builder
      floatingActionButton: FloatingActionButton(
```

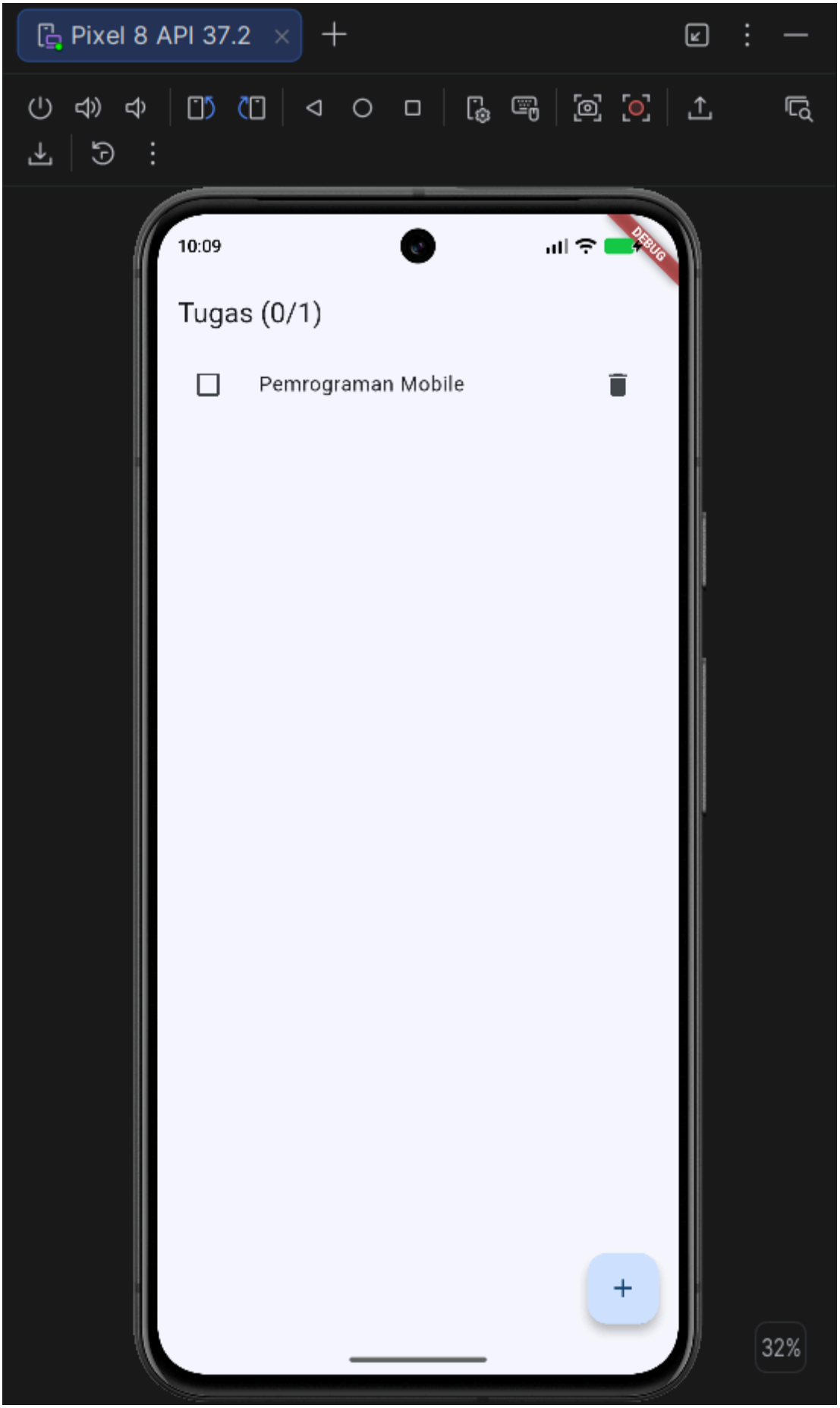
```
        onPressed: () {                                          ✓ 15
          Navigator.push(
            context,
            MaterialPageRoute(builder: (_) => const TambahPage()),
          );
        },
        child: const Icon(Icons.add),
      ), FloatingActionButton
    ); Scaffold
  }
}

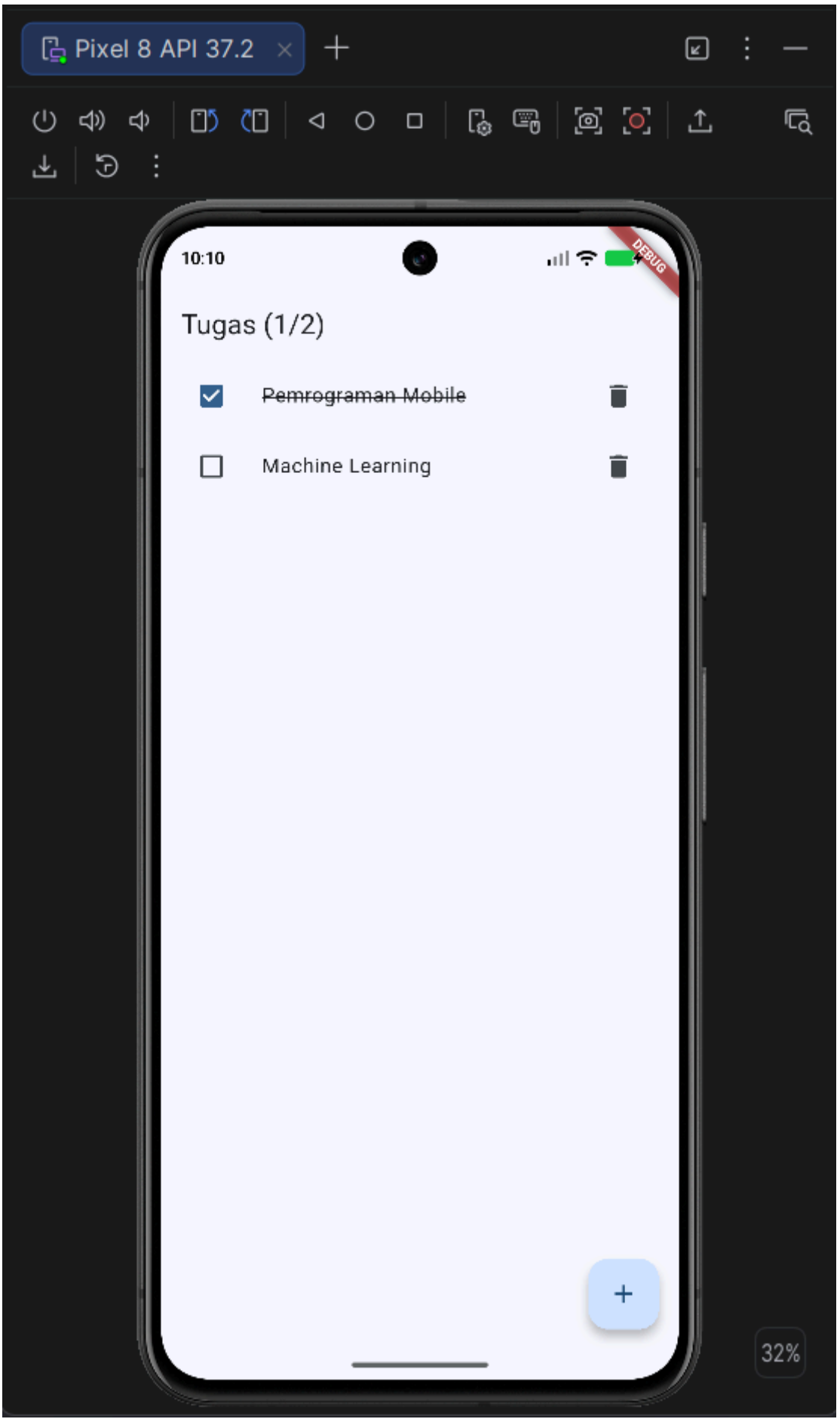
class TambahPage extends StatefulWidget {
  const TambahPage({super.key});
  @override
  State<TambahPage> createState() => _TambahPageState();
}
class _TambahPageState extends State<TambahPage> {
  final _controller = TextEditingController();
  @override
  void dispose() {
    _controller.dispose();
    super.dispose();
  }
  void _simpan() {
    final judul = _controller.text.trim();
    if (judul.isEmpty) return;
    context.read<TugasModel>().tambah(judul);
    Navigator.pop(context);
  }
```

```
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Tambah Tugas')),
      body: Padding(
        padding: const EdgeInsets.all(16),
        child: Column(
          children: [
            TextField(
              controller: _controller,
              autofocus: true,
              decoration: const InputDecoration(
                labelText: 'Judul tugas',
                border: OutlineInputBorder(),
              ), InputDecoration
              onSubmitted: (_) => _simpan(),
            ), TextField
            const SizedBox(height: 12),
            ElevatedButton(onPressed: _simpan, child: const Text('Simpan')),
          ],
        ), Column
      ), Padding
    ); Scaffold
  }
}
```


Pixel 8 API 37.2
10:08
DEBUG
Tugas (0/0)
32%


Pixel 8 API 37.2
10:09
DEBUG
Tambah Tugas
Judul tugas
Pemrograman Mobile
Simpan
32%


Pixel 8 API 37.2
10:09
DEBUG
Tugas (0/1)
Pemrograman Mobile
32%


Pixel 8 API 37.2
10:10
DEBUG
Tugas (1/2)
Pemrograman Mobile
Machine Learning
32%

4. Latihan Mandiri

1. Tambahkan validasi pada TambahPage: judul minimal 3 karakter dan tampilkan pesan error (ubah ke TextFormField + Form).
2. Tambahkan method hapusSelesai() pada TugasModel dan tombol di AppBar untuk menghapus semua tugas yang sudah selesai.
3. Tampilkan SnackBar "Tugas ditambahkan" setelah tugas disimpan.
4. Bila daftar kosong, tampilkan teks "Belum ada tugas" di tengah layar.

main.dart

```dart
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';

class Tugas {
  String judul;
  bool selesai;
  Tugas(this.judul, {this.selesai = false});
}

class TugasModel extends ChangeNotifier {
  final List<Tugas> _items = [];
  List<Tugas> get items => List.unmodifiable(_items);
  int get jumlahSelesai => _items.where((t) => t.selesai).length;

  void tambah(String judul) {
    _items.add(Tugas(judul));
    notifyListeners();
  }

  void toggle(int index) {
    _items[index].selesai = !_items[index].selesai;
    notifyListeners();
  }

  void hapus(int index) {
    _items.removeAt(index);
    notifyListeners();
  }

  void hapusSelesai() {
```

```
    _items.removeWhere((t) => t.selesai);
    notifyListeners();
  }
}

void main() {
  runApp(
    ChangeNotifierProvider(
      create: (_) => TugasModel(),
      child: const MyApp(),
    ), ChangeNotifierProvider
  );
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Daftar Tugas',
      theme: ThemeData(colorSchemeSeed: Colors.blue, useMaterial3: true),
      home: const TugasPage(),
    ); MaterialApp
  }
}

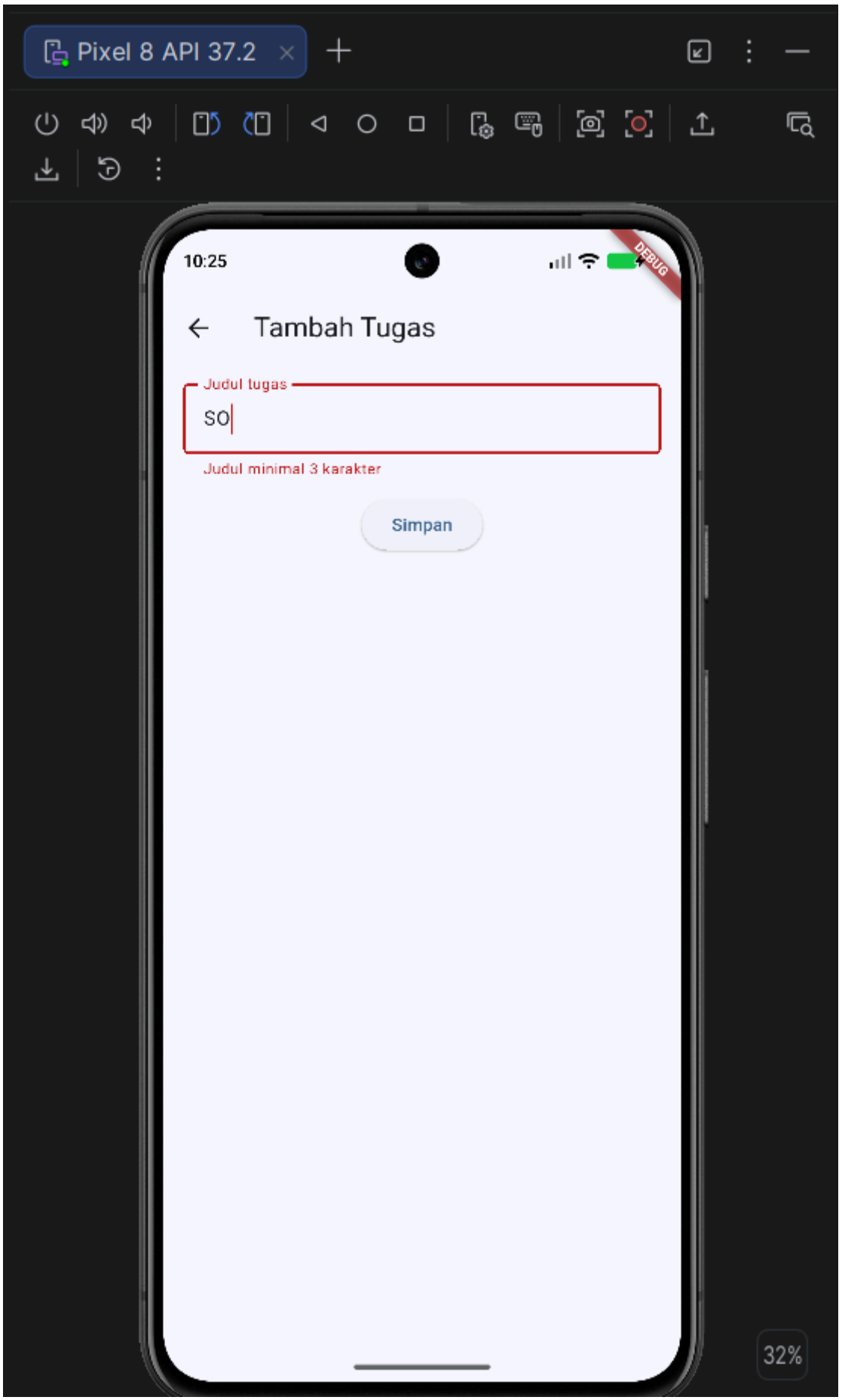
class TugasPage extends StatelessWidget {
  const TugasPage({super.key});
  @override
  Widget build(BuildContext context) {
```

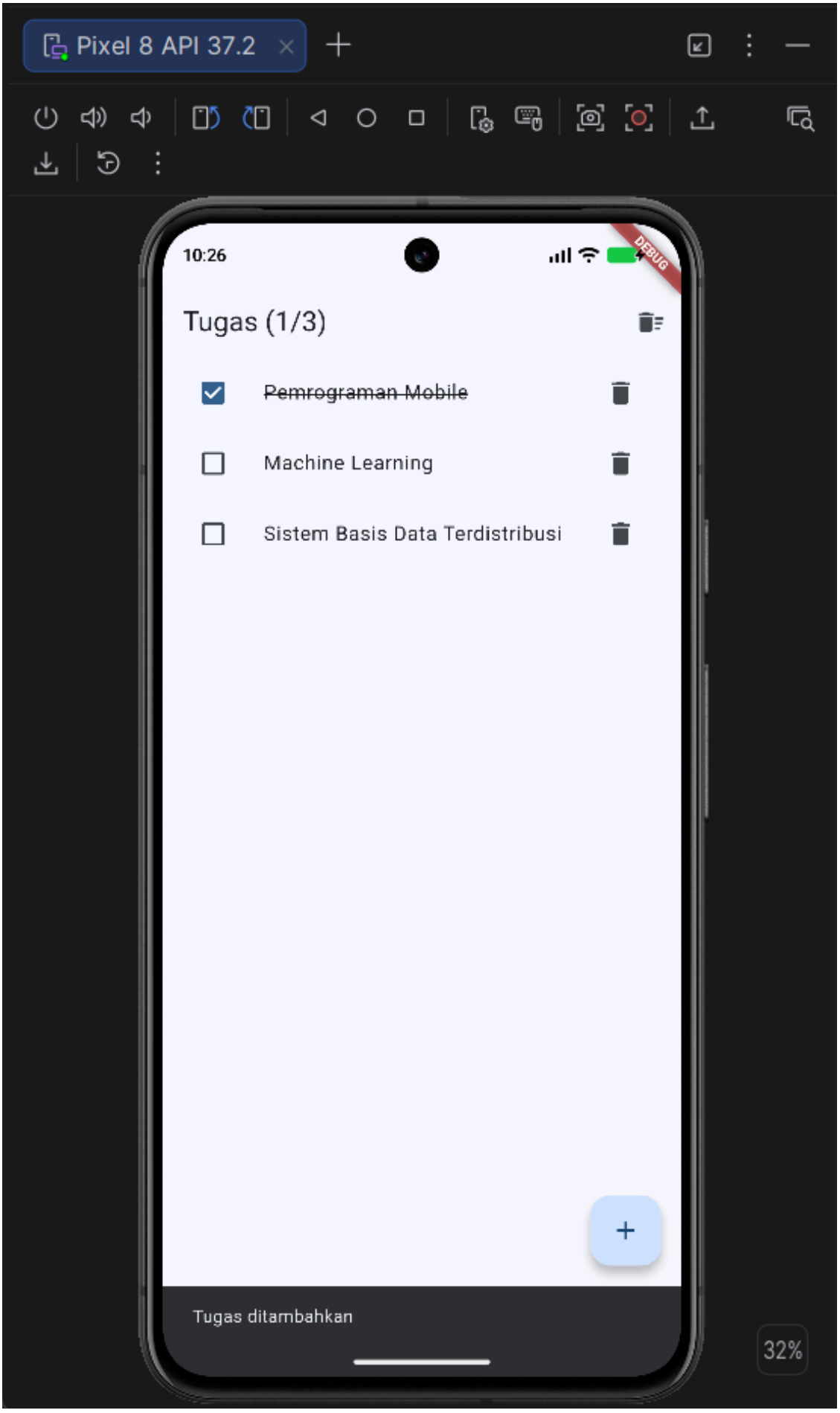
```
    final model = context.watch<TugasModel>();
    return Scaffold(
      appBar: AppBar(
        title: Text('Tugas (${model.jumlahSelesai}/${model.items.length})'),
        actions: [
          IconButton(
            icon: const Icon(Icons.delete_sweep),
            onPressed: () => context.read<TugasModel>().hapusSelesai(),
          ), IconButton
        ],
      ), AppBar
      body: model.items.isEmpty
          ? const Center(child: Text('Belum ada tugas'))
          : ListView.builder(
        itemCount: model.items.length,
        itemBuilder: (context, i) {
          final t = model.items[i];
          return ListTile(
            leading: Checkbox(
              value: t.selesai,
              onChanged: (_) => context.read<TugasModel>().toggle(i),
            ), Checkbox
            title: Text(
```

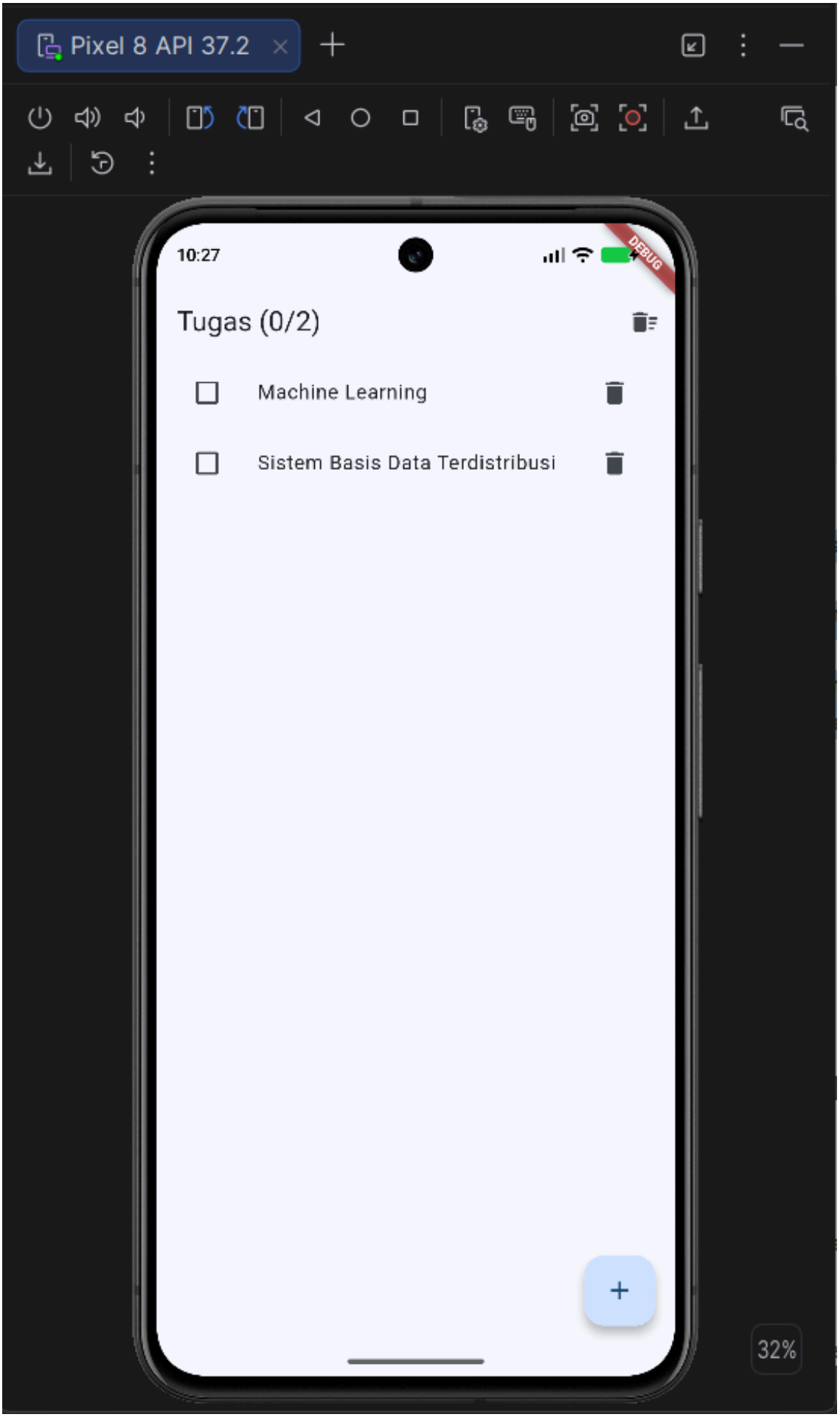
```
83                title: Text(
84                  t.judul,
85                  style: TextStyle(
86                    decoration: t.selesai
87                        ? TextDecoration.lineThrough
88                        : null,
89                  ), TextStyle
90                ), Text
91                trailing: IconButton(
92                  icon: const Icon(Icons.delete),
93                  onPressed: () => context.read<TugasModel>().hapus(i),
94                ), IconButton
95              ); ListTile
96            },
97          ), ListView.builder
98          floatingActionButton: FloatingActionButton(
99            onPressed: () {
100             Navigator.push(
101               context,
102               MaterialPageRoute(builder: (_) => const TambahPage()),
103             );
104           },
105           child: const Icon(Icons.add),
106         ), FloatingActionButton
107       ); Scaffold
108     }
109   }
110
111 class TambahPage extends StatefulWidget {
```

```
112   const TambahPage({super.key});                                    20
113   @override
114   State<TambahPage> createState() => _TambahPageState();
115 }
116
117 class _TambahPageState extends State<TambahPage> {
118   final _formKey = GlobalKey<FormState>();
119   final _controller = TextEditingController();
120
121   @override
122   void dispose() {
123     _controller.dispose();
124     super.dispose();
125   }
```

```
126
127     void _simpan() {
128       if (_formKey.currentState!.validate()) {
129         final judul = _controller.text.trim();
130         context.read<TugasModel>().tambah(judul);
131         ScaffoldMessenger.of(context).showSnackBar(
132           const SnackBar(content: Text('Tugas ditambahkan')),
133         );
134         Navigator.pop(context);
135       }
136     }
137
138     @override
139     Widget build(BuildContext context) {
140       return Scaffold(
141         appBar: AppBar(title: const Text('Tambah Tugas')),
142         body: Padding(
143           padding: const EdgeInsets.all(16),
144           child: Form(
145             key: _formKey,
146             child: Column(
147               children: [
148                 TextFormField(
149                   controller: _controller,
150                   autofocus: true,
151                   decoration: const InputDecoration(
152                     labelText: 'Judul tugas',
153                     border: OutlineInputBorder(),
155                   ), InputDecoration
156                   validator: (v) {
157                     if (v == null || v.trim().length < 3) {
158                       return 'Judul minimal 3 karakter';
159                     }
160                     return null;
161                   },
162                   onFieldSubmitted: (_) => _simpan(),
163                 ), TextFormField
164                 const SizedBox(height: 12),
165                 ElevatedButton(
166                   onPressed: _simpan,
167                   child: const Text('Simpan'),
168                 ), ElevatedButton
169               ],
170             ), Column
171           ), Form
172         ), Padding
173       ); Scaffold
174     }
175
```


Pixel 8 API 37.2
10:25
DEBUG
Tambah Tugas
Judul tugas
SO
Judul minimal 3 karakter
Simpan
32%


Pixel 8 API 37.2
10:26
DEBUG
Tugas (1/3)
Pemrograman Mobile
Machine Learning
Sistem Basis Data Terdistribusi
Tugas ditambahkan
32%


Pixel 8 API 37.2
10:27
DEBUG
Tugas (0/2)
Machine Learning
Sistem Basis Data Terdistribusi
32%


10:27
DEBUG
Tugas (0/0)
Belum ada tugas
32%

5. Tugas

Buat aplikasi Daftar Belanja dengan ketentuan :

- Halaman Form Tambah: nama barang (wajib), jumlah (wajib, angka lebih dari 0), dan kategori (dropdown), dengan validasi pada setiap isian.
- Halaman Daftar: menampilkan semua barang, dapat dicentang sebagai "sudah dibeli" dan dihapus.
- State disimpan pada satu ChangeNotifier dan dibagikan dengan Provider; AppBar menampilkan jumlah barang yang belum dibeli.
- Kumpulkan: tangkapan layar kedua halaman (termasuk tampilan pesan error validasi) dan berkas main.dart (atau tautan repositori).

main.dart

```dart
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';

class Barang {
  String nama;
  int jumlah;
  String kategori;
  bool dibeli;

  Barang(this.nama, this.jumlah, this.kategori, {this.dibeli = false});
}

class BelanjaModel extends ChangeNotifier {
  final List<Barang> _items = [];
  List<Barang> get items => List.unmodifiable(_items);
  int get belumDibeli => _items.where((b) => !b.dibeli).length;

  void tambah(String nama, int jumlah, String kategori) {
    _items.add(Barang(nama, jumlah, kategori));
    notifyListeners();
  }

  void toggle(int index) {
    _items[index].dibeli = !_items[index].dibeli;
    notifyListeners();
  }

  void hapus(int index) {
    _items.removeAt(index);
    notifyListeners();
```

```dart
  }
}

void main() {
  runApp(
    ChangeNotifierProvider(
      create: (_) => BelanjaModel(),
      child: const MyApp(),
    ), ChangeNotifierProvider
  );
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Daftar Belanja',
      theme: ThemeData(colorSchemeSeed: Colors.blue, useMaterial3: true),
      home: const BelanjaPage(),
    ); MaterialApp
  }
}

class BelanjaPage extends StatelessWidget {
  const BelanjaPage({super.key});

  @override
  Widget build(BuildContext context) {
```

```dart
    return Scaffold(
      appBar: AppBar(
        title: Text('Belanja (${model.belumDibeli} belum dibeli)'),
      ), AppBar
      body: model.items.isEmpty
          ? const Center(child: Text('Belum ada barang'))
          : ListView.builder(
        itemCount: model.items.length,
        itemBuilder: (context, i) {
          final b = model.items[i];
          return ListTile(
            leading: Checkbox(
              value: b.dibeli,
              onChanged: (_) => context.read<BelanjaModel>().toggle(i),
            ), Checkbox
            title: Text(
```

```
78                    '${b.nama} (${b.jumlah})',
79                    style: TextStyle(
80                      decoration:
81                      b.dibeli ? TextDecoration.lineThrough : null,
82                    ), TextStyle
83                  ), Text
84                  subtitle: Text(b.kategori),
85                  trailing: IconButton(
86                    icon: const Icon(Icons.delete),
87                    onPressed: () => context.read<BelanjaModel>().hapus(i),
88                  ), IconButton
89                ); ListTile
90              },
91            ), ListView.builder
92            floatingActionButton: FloatingActionButton(
93              onPressed: () {
94                Navigator.push(
95                  context,
96                  MaterialPageRoute(builder: (_) => const TambahBarangPage()),
97                );
98              },
99              child: const Icon(Icons.add),
100           ), FloatingActionButton
101       ); Scaffold
102     }
103   }
104
105 class TambahBarangPage extends StatefulWidget {
```

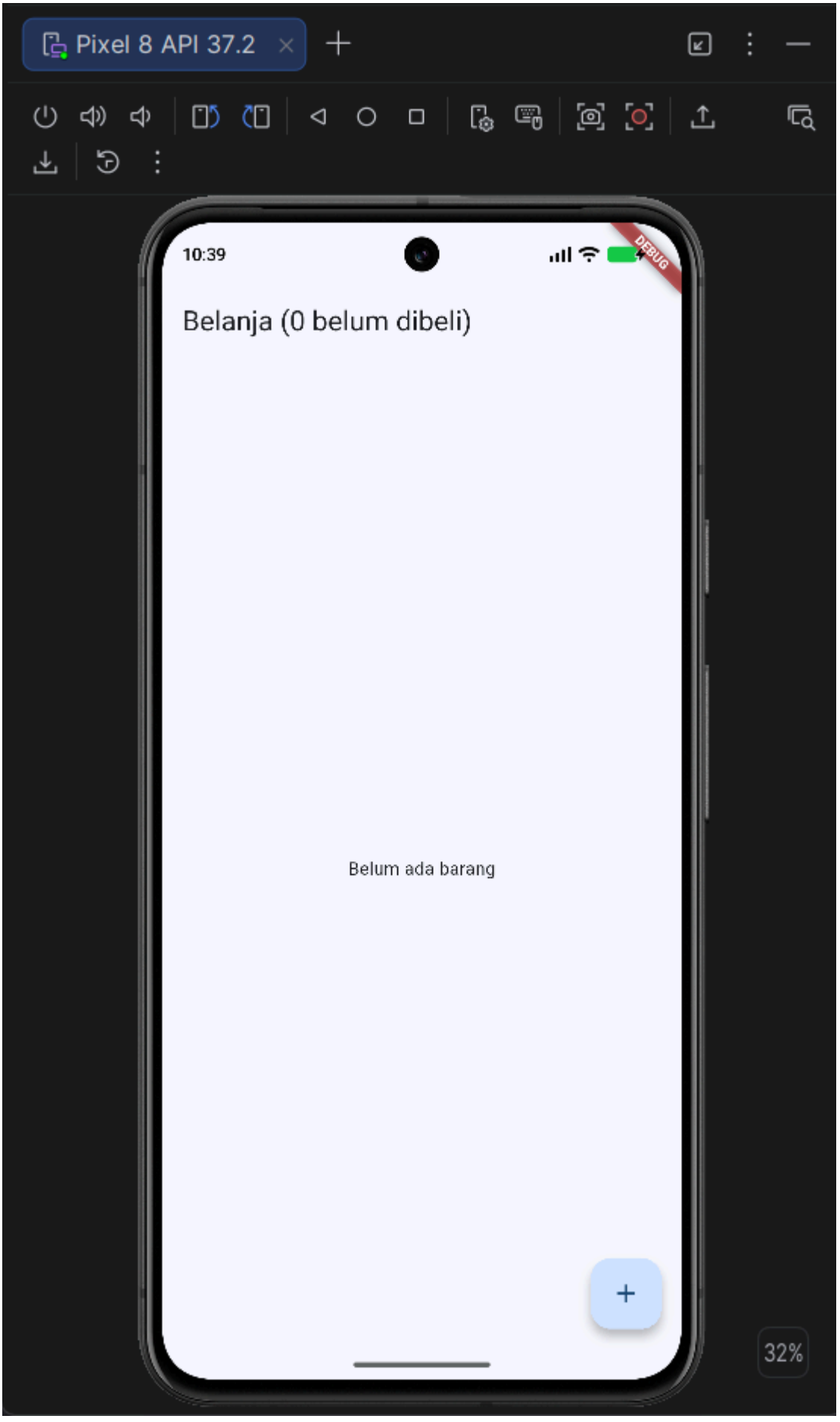
```
    const TambahBarangPage({super.key});                                    42
107
108   @override
109   State<TambahBarangPage> createState() => _TambahBarangPageState();
110 }
111
112 class _TambahBarangPageState extends State<TambahBarangPage> {
113   final _formKey = GlobalKey<FormState>();
114   final _nama = TextEditingController();
115   final _jumlah = TextEditingController();
116   String? _kategori;
117
118   @override
119   void dispose() {
120     _nama.dispose();
121     _jumlah.dispose();
122     super.dispose();
```

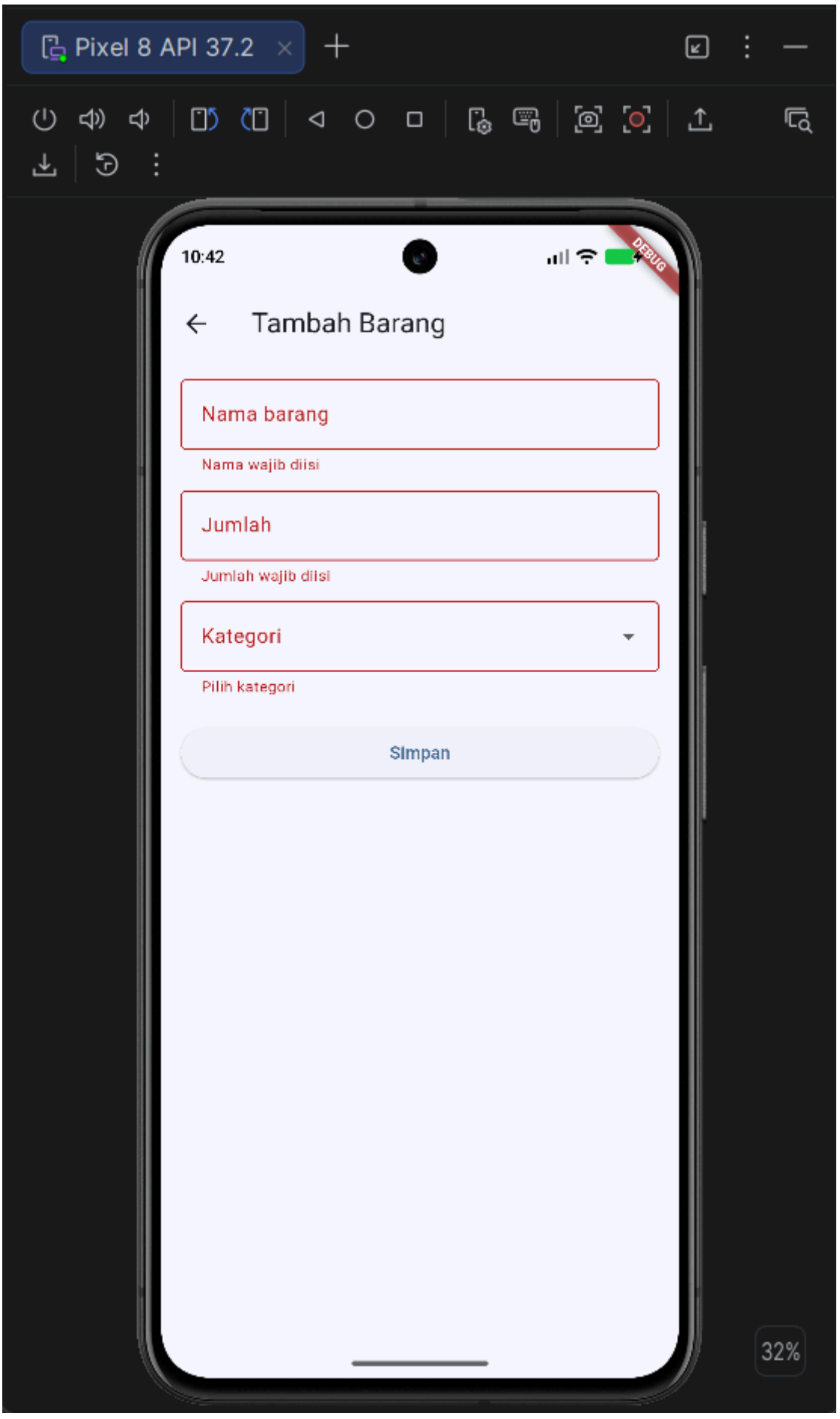
```dart
  }

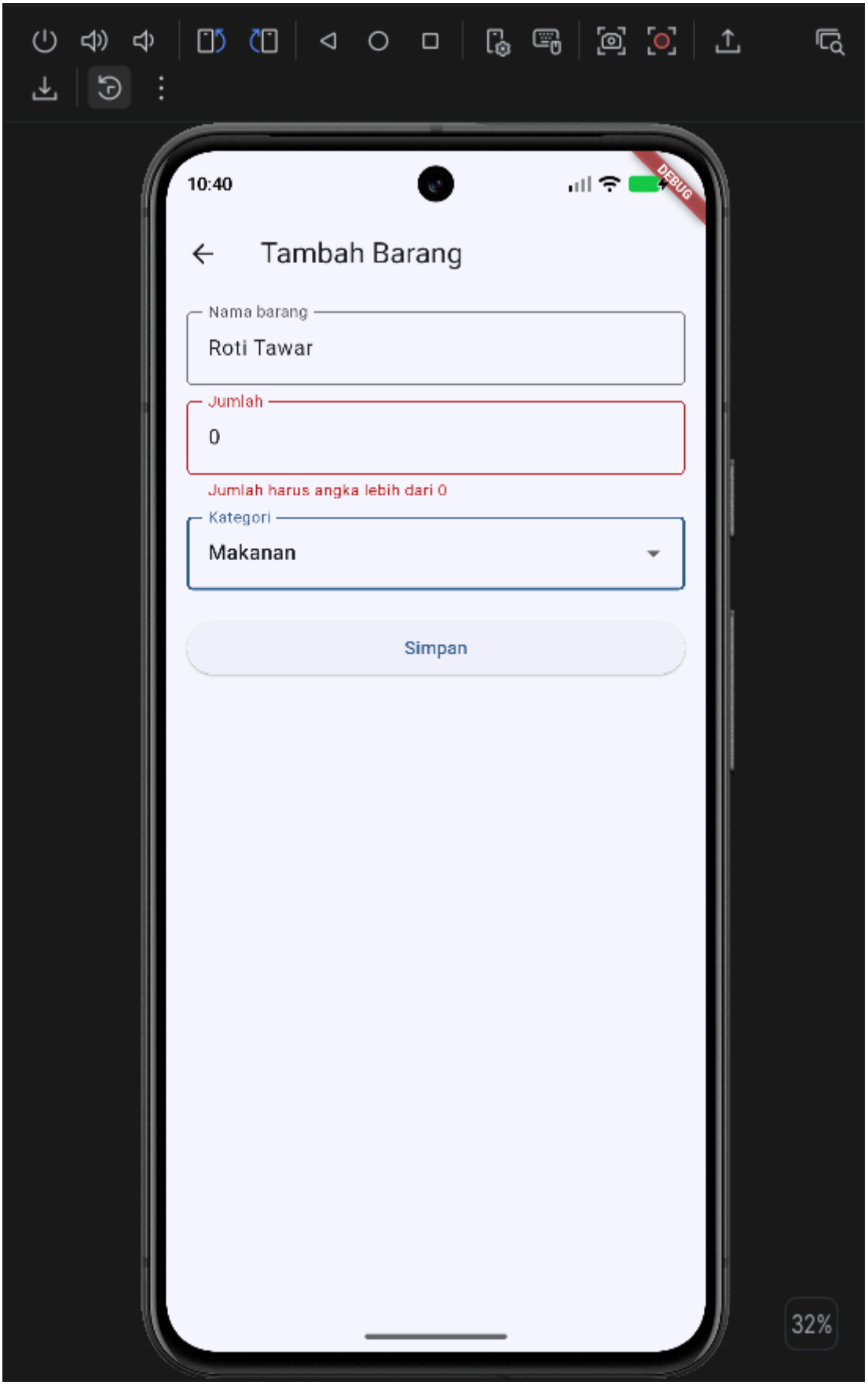
  void _simpan() {
    if (_formKey.currentState!.validate()) {
      context.read<BelanjaModel>().tambah(
        _nama.text.trim(),
        int.parse(_jumlah.text.trim()),
        _kategori!,
      );
      Navigator.pop(context);
    }
  }

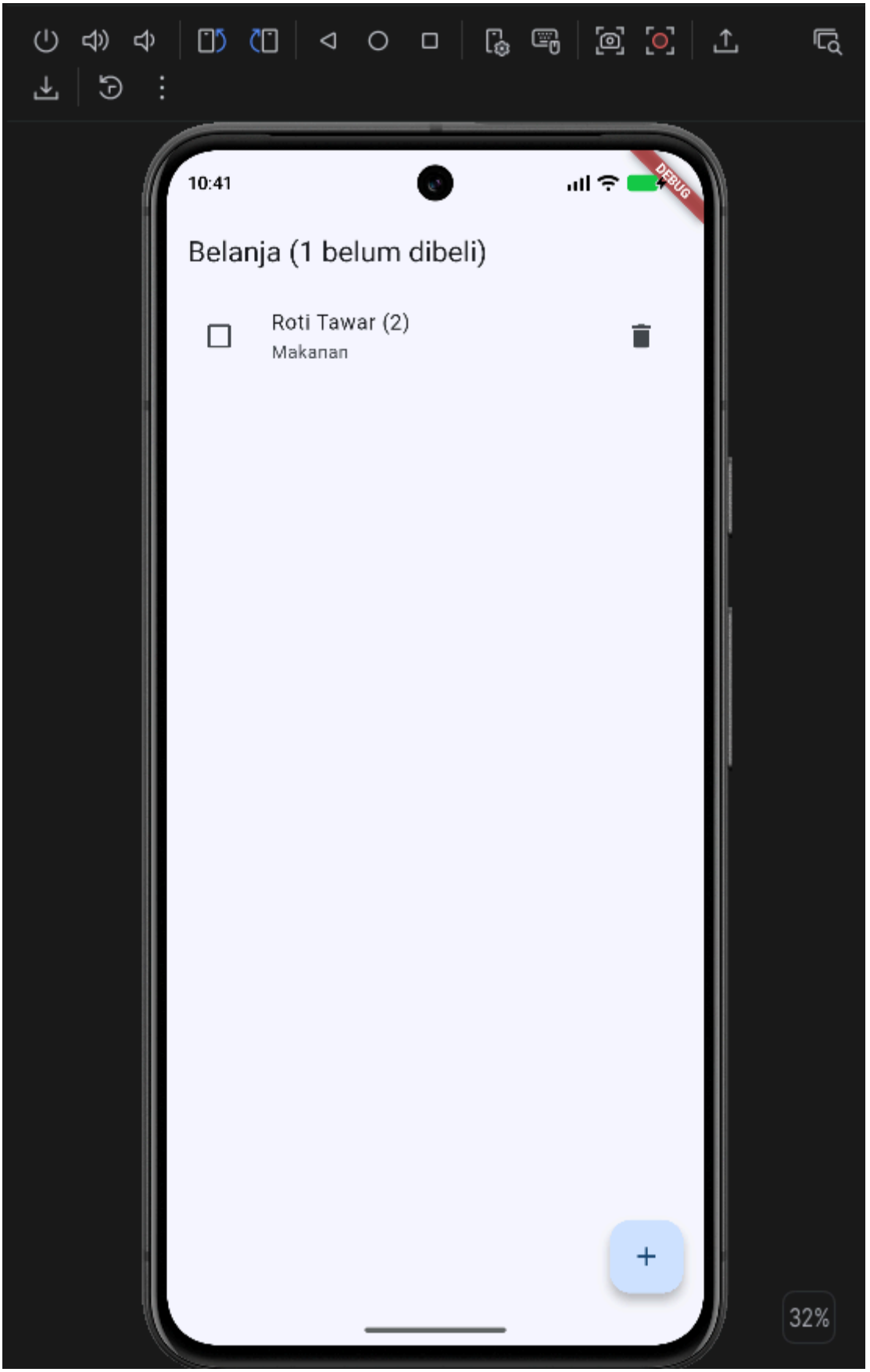
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Tambah Barang')),
      body: Form(
        key: _formKey,
        child: ListView(
          padding: const EdgeInsets.all(16),
          children: [
            TextFormField(
              controller: _nama,
              decoration: const InputDecoration(
                labelText: 'Nama barang',
                border: OutlineInputBorder(),
              ), InputDecoration
              validator: (v) =>                                    42
              (v == null || v.trim().isEmpty) ? 'Nama wajib diisi' : null,
            ), TextFormField
            const SizedBox(height: 12),
            TextFormField(
              controller: _jumlah,
              keyboardType: TextInputType.number,
              decoration: const InputDecoration(
                labelText: 'Jumlah',
                border: OutlineInputBorder(),
              ), InputDecoration
              validator: (v) {
                if (v == null || v.trim().isEmpty) {
                  return 'Jumlah wajib diisi';
                }
                final n = int.tryParse(v.trim());
                if (n == null || n <= 0) {
                  return 'Jumlah harus angka lebih dari 0';
```

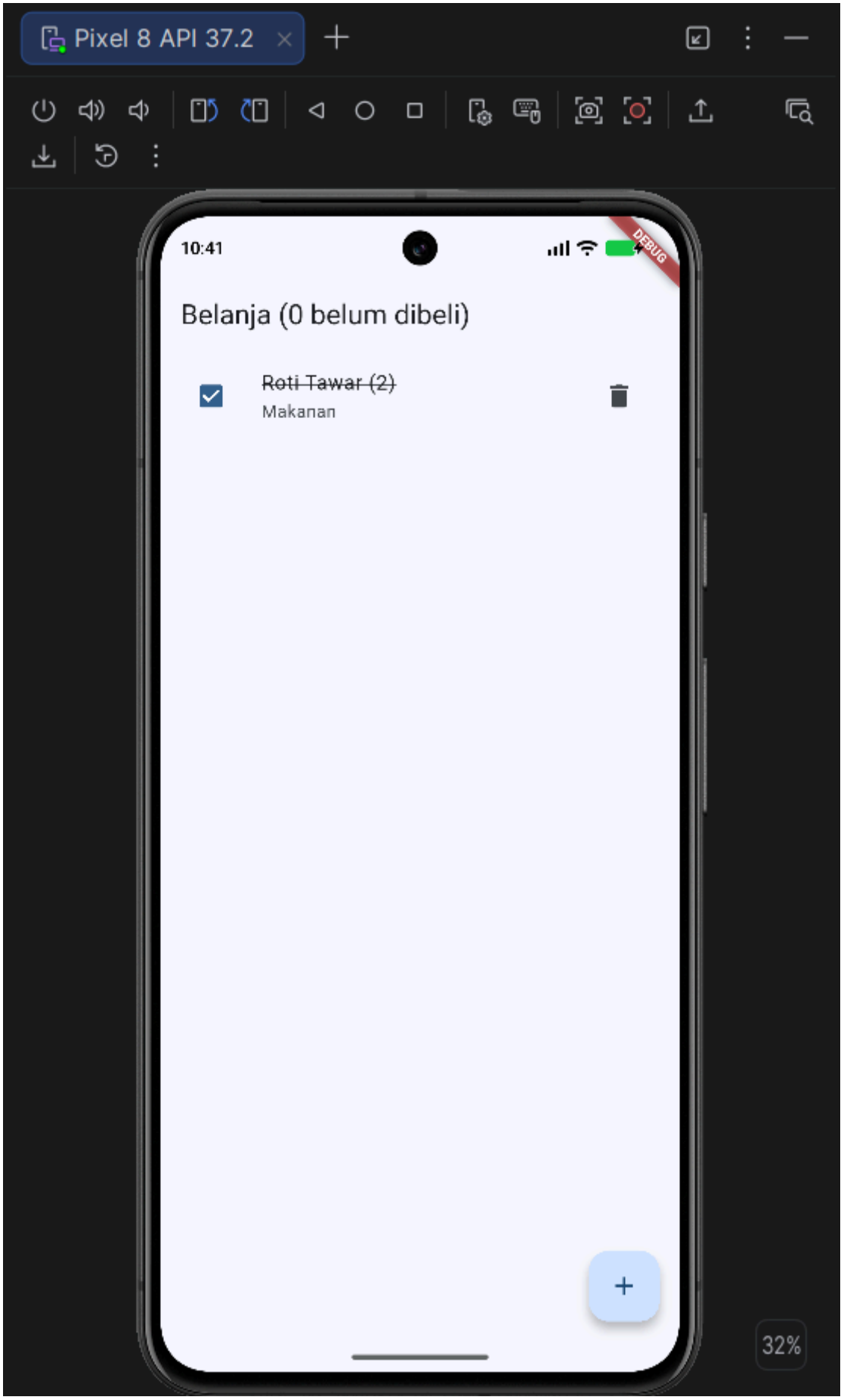
```
                    }
                    return null;
                  },
                ), TextFormField
                const SizedBox(height: 12),
                DropdownButtonFormField<String>(
                  decoration: const InputDecoration(
                    labelText: 'Kategori',
                    border: OutlineInputBorder(),
                  ), InputDecoration
                  items: const [
                    DropdownMenuItem(
                      value: 'Makanan',
                      child: Text('Makanan'),
                    ), DropdownMenuItem
                    DropdownMenuItem(
                      value: 'Minuman',
                      child: Text('Minuman'),
                    ), DropdownMenuItem
                    DropdownMenuItem(
                      value: 'Lainnya',
                      child: Text('Lainnya'),
                    ), DropdownMenuItem
                  ],
                  onChanged: (v) => setState(() => _kategori = v),
                  validator: (v) => v == null ? 'Pilih kategori' : null,
                ), DropdownButtonFormField
                const SizedBox(height: 20),
                ElevatedButton(
                  onPressed: _simpan,
                  child: const Text('Simpan'),
                ), ElevatedButton
              ],
            ), ListView
          ), Form
        ); Scaffold
      }
    }
```


Pixel 8 API 37.2
10:39
DEBUG
Belanja (0 belum dibeli)
Belum ada barang
+
32%


Pixel 8 API 37.2
10:42
DEBUG
Tambah Barang
Nama barang
Nama wajib diisi
Jumlah
Jumlah wajib diisi
Kategori
Pilih kategori
Simpan
32%


10:40
DEBUG
Tambah Barang
Nama barang
Roti Tawar
Jumlah
0
Jumlah harus angka lebih dari 0
Kategori
Makanan
Simpan
32%


10:41
DEBUG
Belanja (1 belum dibeli)
Roti Tawar (2)
Makanan
32%


Pixel 8 API 37.2
10:41
DEBUG
Belanja (0 belum dibeli)
Roti Tawar (2)
Makanan
32%

6. Pertanyaan Refleksi

1. Mengapa TextEditingController harus di-dispose?

TextEditingController harus di-dispose ketika widget sudah tidak digunakan agar resource yang digunakan oleh controller dapat dilepas dan tidak menyebabkan kebocoran memori.

2. Kapan cukup memakai setState, dan kapan sebaiknya beralih ke Provider?

setState() digunakan ketika state hanya diperlukan pada satu widget atau satu halaman. Provider cocok dipakai ketika data yang sama perlu digunakan oleh banyak widget atau halaman.

3. Apa yang terjadi bila notifyListeners() lupa dipanggil? Mengapa?

Jika notifyListeners() lupa dipanggil setelah data berubah, widget yang menggunakan state atau kondisi tersebut tidak diberi tahu bahwa datanya berubah sehingga tampilan dapat tetap menampilkan kondisi lama.

4. Mengapa di dalam onPressed kita memakai context.read, bukan context.watch?

context.read() digunakan karena kita hanya perlu mengambil model untuk menjalankan suatu aksi, tanpa membuat widget tersebut ikut dibangun ulang ketika data berubah. Sementara context.watch() digunakan ketika widget perlu membaca data sekaligus mengikuti perubahan data tersebut.

7. Catatan Pribadi

- TextField : Widget untuk menerima input teks dari pengguna.
- TextEditingController : Mengambil dan mengontrol isi dari TextField.
- dispose() : Melepaskan resource ketika widget sudah tidak digunakan.
- Form : Widget untuk mengelompokkan input yang membutuhkan validasi.
- TextFormField : TextField yang dapat menggunakan fungsi validasi.
- validator : Digunakan untuk memeriksa apakah input valid atau tidak.
- GlobalKey<FormState> : Digunakan untuk menjalankan validasi form.
- validate() : Menjalankan semua fungsi validator pada form.
- DropdownButtonFormField : Input berbentuk pilihan / dropdown.
- CheckboxListTile : Input berupa checkbox dengan teks.
- SnackBar : Menampilkan pesan singkat pada bagian bawah layar.
- ChangeNotifier : Untuk menyimpan state dan memberi tahu widget data berubah.

- Provider : Untuk membagikan state kepada widget atau halaman yang membutuhkan.
- notifyListeners() : Memberitahu widget yang berlangganan bahwa state telah berubah.
- context.watch() : Membaca state dan membangun ulang widget ketika state berubah.
- context.read() : Membaca state tanpa membangun ulang widget ketika state berubah.